Heredity and Evolution

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🧬 1. Heredity and Hereditary Changes

🔑 What is Heredity?

- **Heredity** = Transfer of biological characters from one generation to another **via genes**.
- **Gregor Johann Mendel** is the pioneer of modern genetics.

🕰️ Historical Timeline of Key Discoveries

1901 ──► Hugo de Vries proposes the Mutational Theory
(explains reasons behind sudden changes)

1902 ──► Walter Sutton observes paired chromosomes
in grasshopper cells (first observation of
chromosomal pairs)

1944 ──► Oswald Avery + Mclyn McCarty + Colin McLeod
PROVE: DNA is the genetic material
(in all living organisms except viruses)

1961 ──► François Jacob + Jacques Monod (French)
Propose model for protein synthesis via DNA
in bacterial cells → unlocks genetic codes

↓
Recombinant DNA Technology emerges
↓
Birth of Genetic Engineering

💡 Applications of Heredity Science

- Diagnosis, treatment, and prevention of **hereditary disorders**
- Production of **hybrid varieties** of animals and plants
- Industrial processes using **microbes**

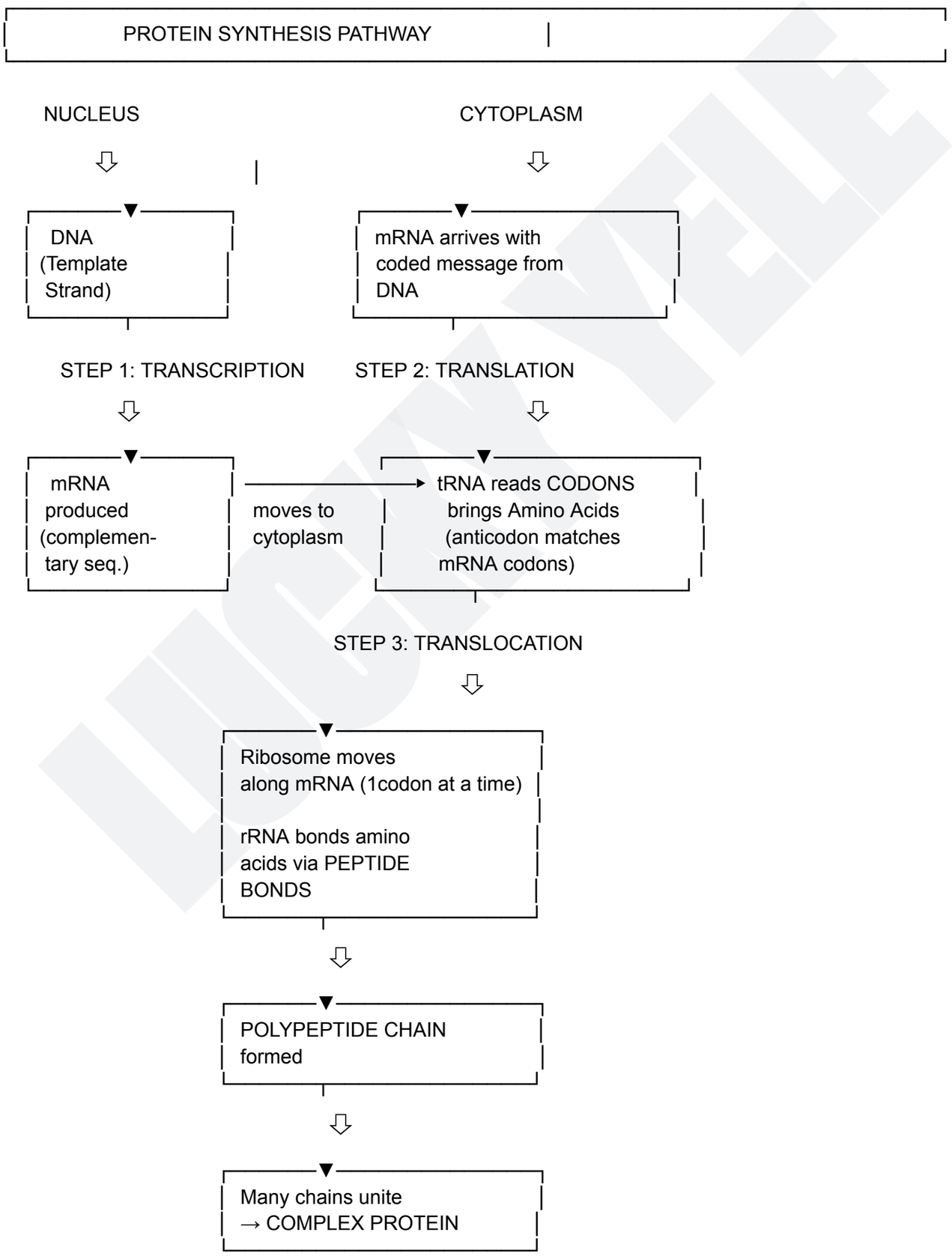
⚙️ 2. Transcription, Translation & Translocation

🔑 The Central Dogma

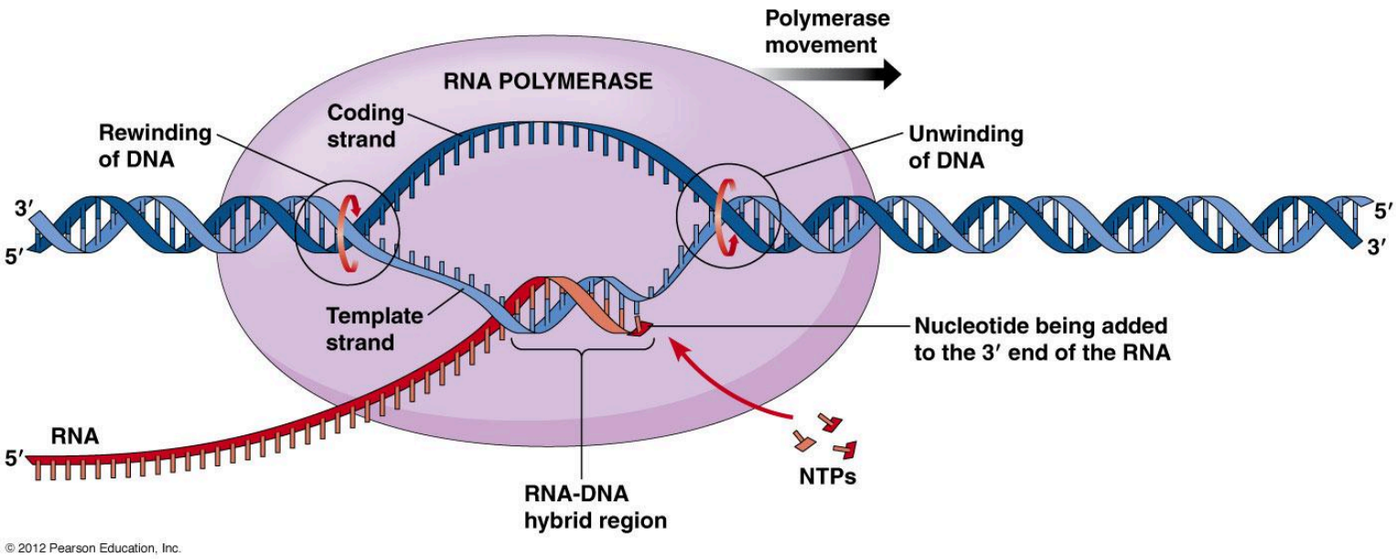
Central Dogma of Molecular Biology: DNA ➡ RNA ➡ Protein

- Information about **protein synthesis** is stored in DNA.
- Proteins are synthesized by DNA **through RNA**.
- These proteins control the **structure and functioning** of the body.

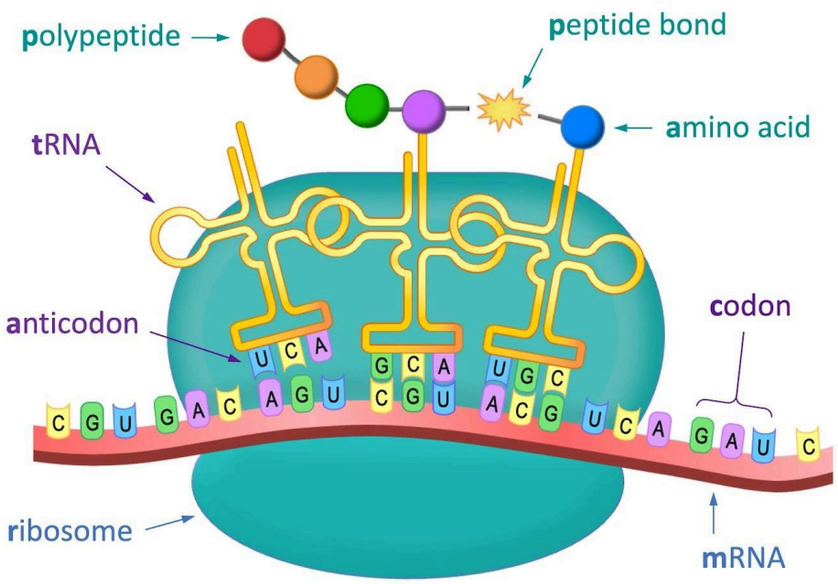
📊 Three Stages of Protein Synthesis



TRANSCRIPTION :



TRANSLATION AND TRANSLOCATION:



📋 Comparison Table: Transcription vs Translation vs Translocation

Feature	Transcription	Translation	Translocation
Definition	Synthesis of mRNA from DNA	Reading mRNA to assemble amino acids	Ribosome movement along mRNA
Location	Nucleus	Cytoplasm (Ribosome)	Cytoplasm (Ribosome)
Key Molecule Made	mRNA	Polypeptide chain	— (Process of movement)
Template Used	One strand of DNA	mRNA codons	mRNA
Key Players	DNA, RNA Polymerase	mRNA, tRNA, rRNA	Ribosome
Unit of Code	Nucleotide sequence	Triplet codon	One codon distance

🔬 Key Definitions

- **Triplet Codon** = A code of **3 nucleotides** on mRNA that represents one amino acid.
- **Anticodon** = Complementary sequence on tRNA that matches the codon on mRNA.
- **Peptide Bond** = The bond linking amino acids together (formed with help of **rRNA**).

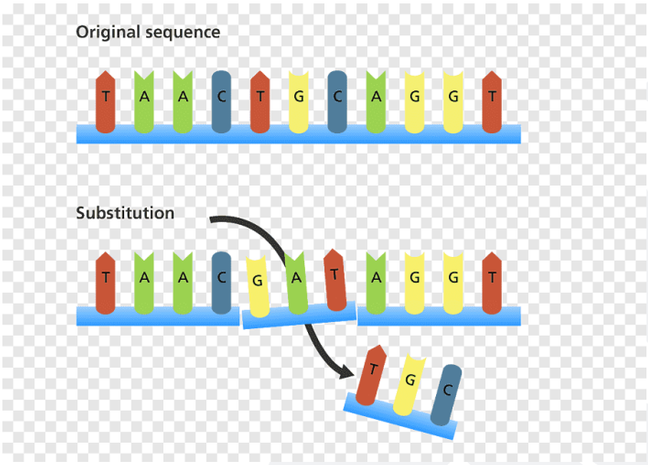
🏆 Scientist Spotlight

> 🌟 **Dr. Har Govind Khorana** (Indian-origin scientist) Discovered **triplet codons for all 20 amino acids.** Awarded the **Nobel Prize in 1968** (shared with two other scientists).

⚠️ 3. Mutation

🔑 What is Mutation?

- Genes are **normally transmitted unchanged** to offspring → ensuring parental characters pass on.
- However, sometimes **sudden changes** occur in genes.
- When **any nucleotide changes its position** in a gene → causes a change = **Mutation**.



📊 Types of Impact

Type	Description	Example
Minor Mutation	Small positional change in nucleotide	Slight variation in a trait
Considerable Mutation	Significant change in gene sequence	**Sickle Cell Anaemia**

> 🔑 **Key Principle:** Mutations are the raw material for evolution — they introduce the variation that natural selection acts upon.

🌍 4. Evolution

🔑 Definition

> **Evolution** = The gradual, slow change occurring in living organisms over a **very long duration**, leading to the formation of new species through **natural selection**.

🌱 Origin of Life — Step-by-Step Flow

3.5 Billion Years Ago: No life on Earth
↓
Simple elements present in ocean
↓
Simple organic + inorganic compounds formed
↓
Complex compounds formed
(Proteins + Nucleic Acids)
↓
First PRIMITIVE CELLS formed
(from mixture of organic + inorganic compounds)
↓
Cell numbers increased
(at cost of surrounding chemicals)
↓
Differences appeared among cells
↓
Well-adapted cells: SURVIVED & grew
↓
CRORES OF DIVERSE SPECIES today
(from unicellular Amoeba → Man → Giant Whale)

Poorly-adapted cells: PERISHED (could not adjust to surroundings)

📊 Biodiversity Spectrum (Per Source Text)

Kingdom	Smallest / Simplest	Largest / Most Complex
Animalia	Unicellular Amoeba, Paramecium	Man, Giant Whale
Plantae	Unicellular Chlorella	Huge Banyan Tree

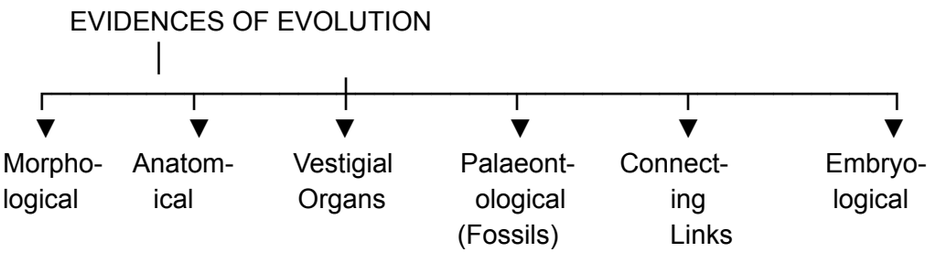
- Life exists **everywhere** — equator to both poles; in air, water, land, and rock.

📜 Theory of Evolution (Key Points)

1. First living material **(protoplasm)** formed in the **ocean**.
2. Unicellular organisms formed first.
3. Gradually, changes produced **larger and more complex organisms**.
4. All changes were **slow and gradual** — spanning up to **300 crore years**.
5. Changes were **multi-dimensional** → led to diverse types of organisms.

> **Progressive development of plants and animals from ancestors with different structural and functional organization is called Evolution.**

🔍 5. Evidences of Evolution



🦴 Evidence 1: Morphological Evidences

- **Morphological** = Related to external shape/structure.
- **In Animals:** Similarities in mouth structure, eye position, nostril structure, ear pinnae, and body hair distribution are observed across groups.
- **In Plants:** Similarities in leaf shape, leaf venation, leaf petiole.
- **Conclusion:** These similarities indicate a **common ancestry** among these groups.



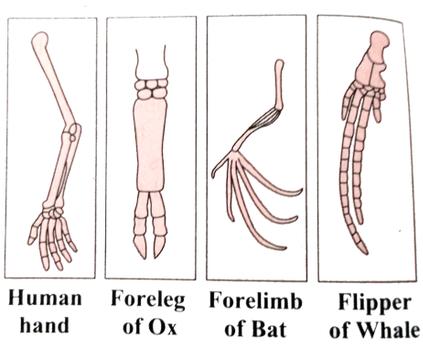
🦴 Evidence 2: Anatomical Evidences

- **Anatomical** = Related to internal structure of organs.
- Though the following organs look **completely different** externally and serve **different functions**, their **internal bone structure and joints are similar**:

Organ	Animal	Function
Hand	Human	Grasping, manipulating
Foreleg	Ox	Walking, weight-bearing
Flipper	Whale	Swimming
Patagium (wing)	Bat	Flying

- **Conclusion:** Similar bone structure = **Common ancestor** for all these animals.

🔑 These are called **Homologous Organs** — same basic structure, different functions.



🦴 Evidence 3: Vestigial Organs

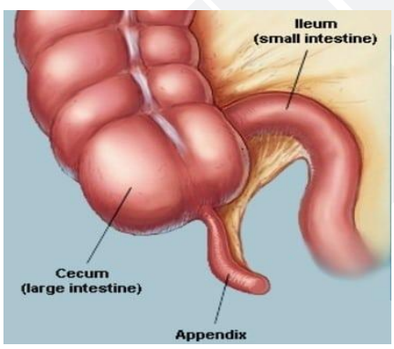
- **Vestigial Organs** = Degenerated or underdeveloped **useless organs** of organisms.

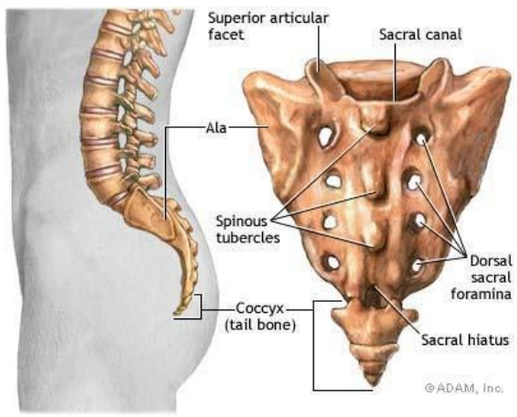
🔄 Why do Vestigial Organs Exist?

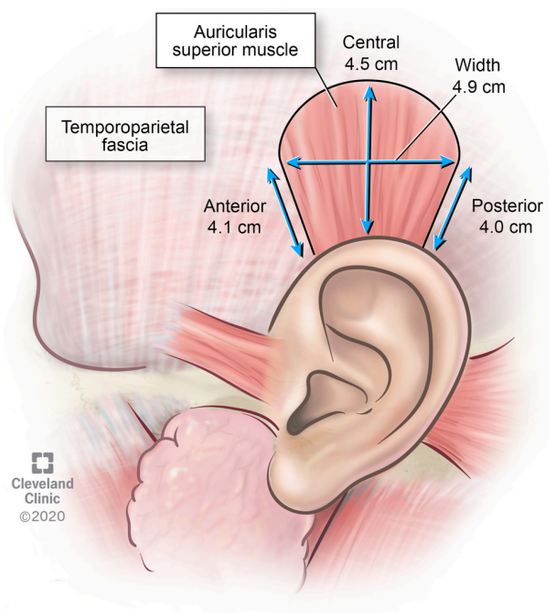
Organ USEFUL in certain environment
↓
Environment CHANGES
↓
Organ becomes USELESS or even HARMFUL
↓
Organ begins to DEGENERATE
(as per natural selection principle)
↓
Takes THOUSANDS OF YEARS to disappear
↓
Seen in DIFFERENT PHASES of disappearance
in different animals

📊 Vestigial Organs Comparison Table

Vestigial Organ	Organism (Non-functional)	Organism (Functional)
Appendix	Human	Ruminants
Ear Pinna Muscles	Human	Monkeys (to move pinna)
Tail Bone (Coccyx)	Human	Animals with tails
Wisdom Teeth	Human	Ancestors (for raw food)
Body Hairs	Human	Fur-bearing mammals







🦴 Evidence 4: Palaeontological Evidences (Fossils)

What Are Fossils?

- **Fossils = Remnants and impressions of organisms preserved underground**
- **Formed when organisms get buried due to natural disasters — floods, earthquakes, volcanoes**
- **The buried remains are preserved in sedimentary rock layers over millions of years**
- **Palaeontology = the branch of science that studies fossils**
- **Study of fossils is a critical aspect of understanding evolution**

☢️ Carbon Dating — How We Determine Age of Fossils

Concept	Detail
What stops at death?	Carbon consumption by plants/animals
Radioactive Carbon	C-14 (Carbon-14)
Non-radioactive Carbon	C-12 (Carbon-12)
Post-mortem Change	The C-14 : C-12 ratio changes continuously as C-14 decays while C-12 remains constant
Measurement	Radioactivity of C-14 and the C-14 to C-12 ratio
Resulting Data	Time elapsed since death (age of the fossil)
Applications	Palaeontology & Anthropology — aging fossils and ancient manuscripts

💡 **Key Logic: While alive → organism maintains constant C-14:C-12 ratio. After death → C-14 keeps decaying, C-12 doesn't → ratio keeps changing → the *amount* of change tells us *how long* the organism has been dead.**

Reading the Fossil Diagram (Sedimentary Rock Layers)

- **Deeper = Older | Shallower = Newer (see the arrow on the left)**
- **Sediment layers trap organisms in sequence — bottom layers formed first**

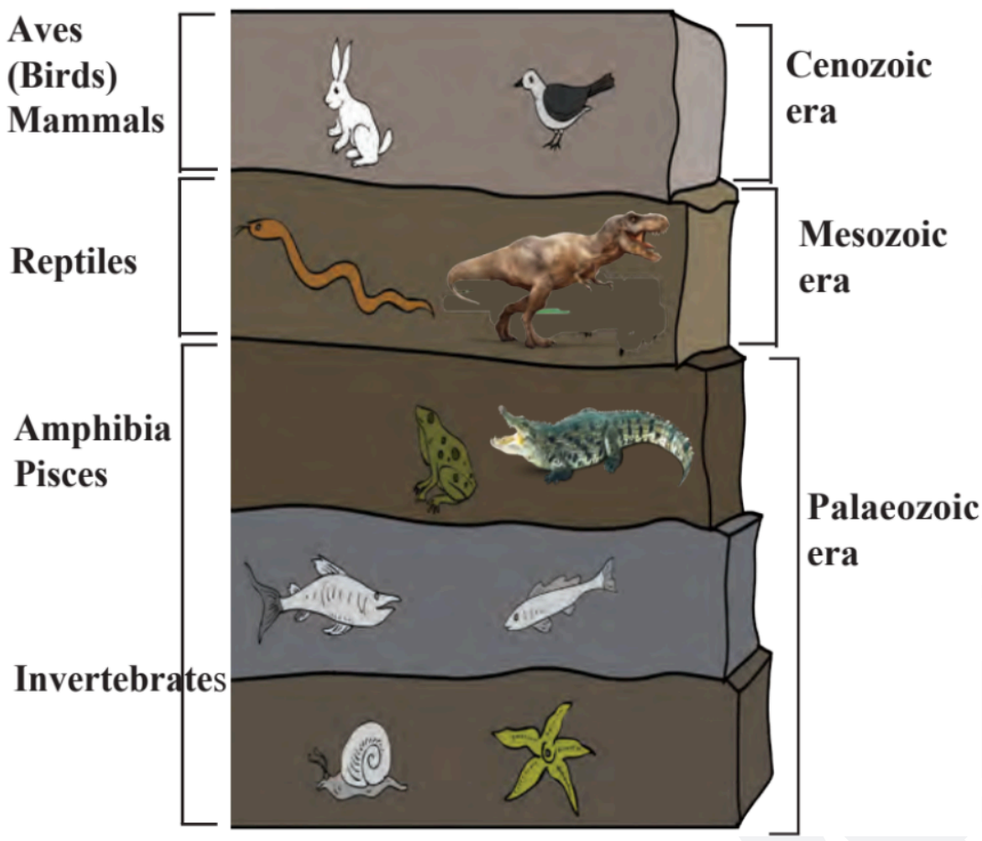
Layer (Bottom → Top)	Geological Era	Organisms Found
🟫 Deepest layer	Palaeozoic era	Invertebrates — snails, starfish (no backbone)
🟫 3rd layer	Palaeozoic era	Pisces (fish) + Amphibia (frogs)
🟤 2nd layer	Mesozoic era	Reptiles — snakes, dinosaurs
🟡 Top layer	Cenozoic era	Aves (birds) + Mammals

🔁 The Grand Evolutionary Conclusion

- **Invertebrates appear first (deepest) → Vertebrates appear later (higher layers)**
- **This rock-solid fossil sequence shows that:**

  🧬 **"Vertebrates have slowly originated from Invertebrates"**

- **The evolutionary order confirmed by fossils: Invertebrates → Pisces → Amphibia → Reptiles → Aves + Mammals**



🦴 Evidence 5: Connecting Links 🔗

- **Connecting Links** = Plants/animals showing characters of **two different groups**, placing them between two groups.

📊 Connecting Links Table

Organism	Characters of Group A	Characters of Group B	Links Between
Peripatus	Segmented body, thin cuticle, parapodia-like organs (Annelida)	Tracheal respiration, open circulatory system (Arthropoda)	Annelida & Arthropoda
Duck-billed Platypus	Lays eggs (Reptiles)	Mammary glands, hairs (Mammals)	Reptiles & Mammals
Lung Fish	Is a fish	Breathes using lungs (Amphibians)	Fish & Amphibians

- **Conclusion:** Mammals evolved from reptiles; Amphibians evolved from fishes.



Duck billed platypus



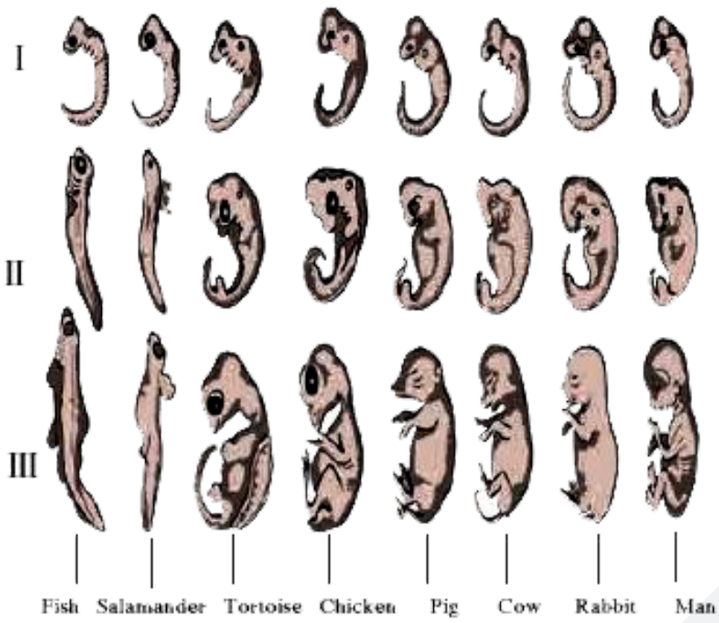
Lungfish



Peripatus

🦴 Evidence 6: Embryological Evidences 🥚

- Comparative study of **embryonic developmental stages** of various vertebrates shows:
  - **Initial stages** → All embryos show **extreme similarities**.
  - Similarities **decrease gradually** as development progresses.
- **Conclusion:** Similarities in early embryonic stages indicate a **common origin** for all vertebrate animals.



Embryos during different stages

🌿 6. Darwin's Theory of Natural Selection

🔑 Background

- **Charles Robert Darwin (1809–1882)** — English biologist.
- Collected **innumerable specimens** of plants and animals.
- Published his findings in the book: **"Origin of Species"**.

📊 Darwin's Theory — Step by Step

All organisms REPRODUCE PROLIFICALLY
(produce large numbers of offspring)
↓
All organisms COMPETE with each other
(life-threatening competition)
↓
Only organisms with USEFUL MODIFICATIONS
survive the competition
↓
NATURE SELECTS organisms fit to live
(Unfit organisms PERISH)
↓
Surviving organisms REPRODUCE
↓
NEW SPECIES formed with their own
specific characters

Core Principle of Darwinism: "Survival of the Fittest" Nature selects only those organisms which are fit to live; the rest perish.

🦋 Example 1 — The Peppered Moth (*Biston betularia*)

This serves as the most **iconic real-world proof** of Darwin's Theory of Natural Selection.

🏭 Pre-Industrial England (Before 1800s)



- Tree bark was primarily covered in **light-colored lichen**.
- Moth populations existed in **two distinct forms**: light-colored (peppered) and dark (melanic).
- **Light moths** were perfectly **camouflaged** against lichens → birds could not spot them → they **survived and reproduced**.
- **Dark moths** remained highly **visible** → predated upon by birds → they **perished before reproducing**.
- Result: The **light-colored moths dominated** the overall population.

🏭 Industrial Revolution (1800s–1900s)

- Industrial soot killed lichens → tree bark turned **dark (blackened)**.
- In this changed environment, **dark moths** became **camouflaged** → they survived to breed.
- **Light moths** became clearly **visible** → easily eaten by predatory birds.
- Result: **Dark moths dominated** the population (reaching 95% in industrial zones by 1895).

🐦 Example 2 — Darwin's Finches (*Geospiza* spp.)

🌍 The Setup

Darwin visited the **Galápagos Islands** (14 islands, each slightly different). He found **13–14 species of finches**, all descended from **ONE common ancestor** that flew from South America.

🔑 The Key Observation — Beak Variation

Island Food Source	Beak Shape	Finch Type
Hard seeds	**Thick, crushing beak**	Ground finch
Insects in bark	**Long, probing beak**	Woodpecker finch
Cactus nectar	**Long, curved beak**	Cactus finch
Small soft seeds	**Thin, pointed beak**	Warbler finch

⚙️ How Natural Selection Worked Here

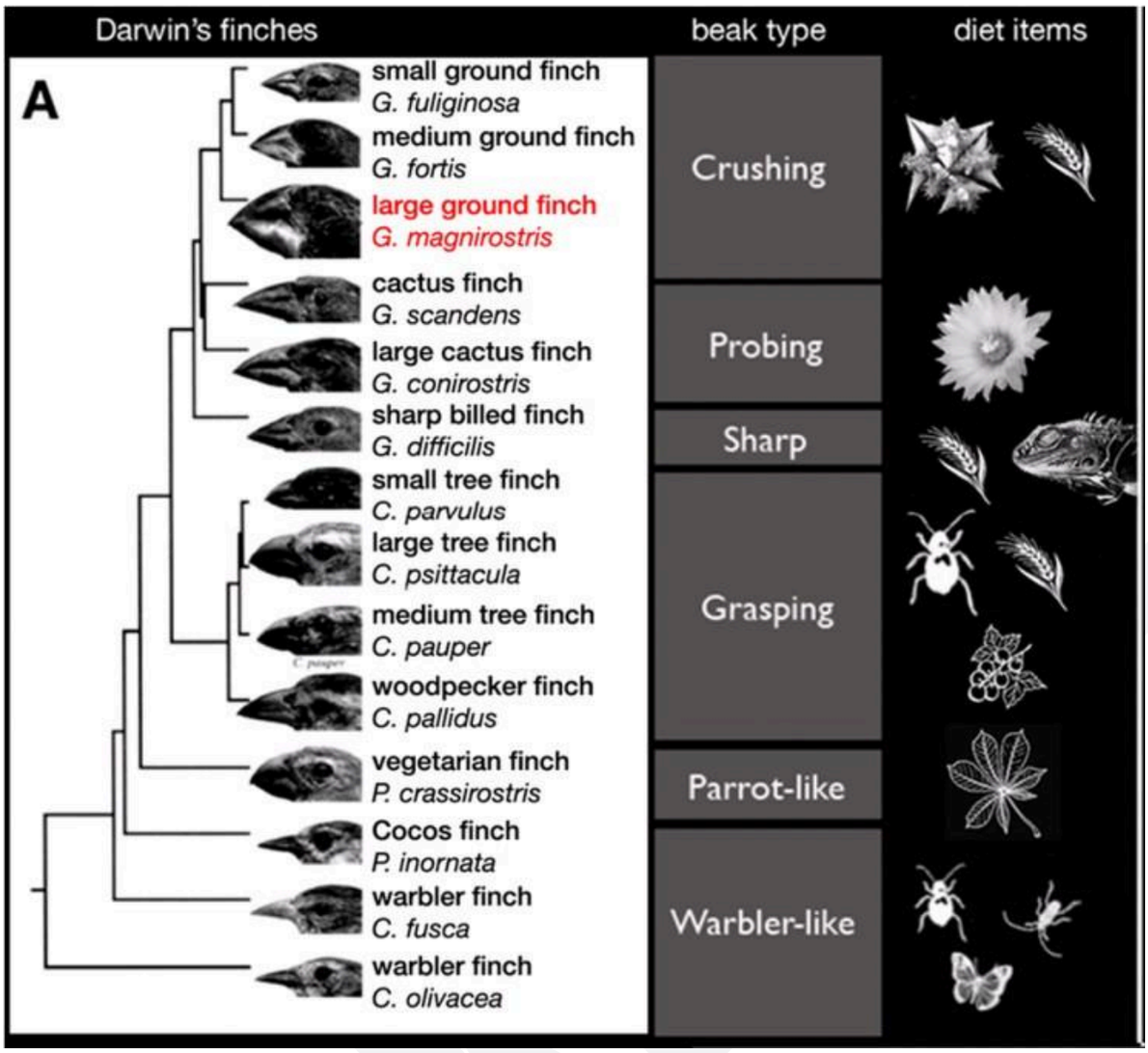
1. Original finches landed on islands → **different islands had different food**
2. Birds with beaks **better suited** to available food **survived longer**
3. They **reproduced more** → passed beak traits to offspring
4. Over **thousands of generations** → beak shapes **diverged** per island
5. Eventually, populations became **reproductively isolated** → **new species** formed

🧬 This process is called **Adaptive Radiation** — one ancestor species **rapidly diversifies** into many species to fill different ecological niches.

⚠️ Objections to Darwin's Theory

No.	Objection
1	Natural selection is **not the only factor** responsible for evolution.
2	Darwin gave **no explanation** about what makes modifications useful or useless.
3	**No explanation** about slow (gradual) vs abrupt (sudden) changes.

Despite these objections, **Darwin's work on evolution has been a milestone** in the history of science.



🦒 7. Lamarckism

🔑 Who was Lamarck?

- **Jean-Baptiste Lamarck (1744–1829)** — French naturalist.

📜 Core Idea of Lamarckism

Principle of 'Use or Disuse of Organs': Morphological changes occurring in living organisms are caused by their **activities (use)** or **inactivity (disuse)**, and these acquired characters are **inherited by the next generation**. This is called **"Theory of Inheritance of Acquired Characters" = Lamarckism.**

📊 Lamarck's Examples

Organism	Activity / Cause	Result (Acquired Character)
Giraffe	Extended neck for generations to browse tall plants	Neck became very **long**
Ironsmith	Frequent hammering movements	Shoulders became **very strong**
Ostrich / Emu	No use of wings	Wings became **weak**
Swan / Duck	Living in water, legs used for swimming	Legs became **suited for swimming**

Organism	Activity / Cause	Result (Acquired Character)
Snake	Body modifications for burrowing habit	**Lost their legs**

⚠️ Criticism of Lamarckism

Was ACCEPTED: Development of organs due to use / degeneration due to disuse ✅

Was REJECTED: Transfer of acquired characters to next generation ❌

- It was **verified many times** that modifications acquired during one's lifetime are **NOT transferred** to the next generation.
- Therefore, **Lamarck's theory was disproved**.

📊 Darwin vs Lamarck — Comparison Table

Feature	🔵 Darwinism	🟠 Lamarckism
Proposed by	Charles Darwin (1809–1882)	Jean-Baptiste Lamarck (1744–1829)
Core Mechanism	Natural Selection (Survival of the Fittest)	Use or Disuse of Organs
Role of Environment	Selects pre-existing variations	Causes new modifications directly
Inheritance Claim	Naturally selected traits passed on	Acquired characters transferred
Giraffe Neck Explanation	Those with longer necks survived and reproduced	Giraffe stretched neck → neck grew longer
Current Status	Widely accepted (with modifications)	Largely disproved

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🌱 8. Speciation

🔑 Definition

Speciation = Formation of new species of plants and animals as an effect of evolution.

🔑 What is a Species?

A **species** = A group of organisms that can produce **fertile individuals** through natural reproduction.

📜 Factors Responsible for Speciation

GENETIC VARIATION
+
GEOGRAPHICAL CHANGES
+
REPRODUCTIVE CHANGES
+
GEOGRAPHICAL or REPRODUCTIVE ISOLATION

⇩

SPECIATION (Formation of New Species)

- Each species grows in **specific geographical conditions**.
- Their food, habitat, reproductive ability, and period are **different**.

🔷 Core Causes of Speciation

1. 🌍 Geographical Isolation (Allopatric Speciation)

- A **physical barrier** (mountain, river, ocean, desert) **separates a population** into two groups.
- Over thousands of generations, each group **evolves independently** due to:
  - Different food sources
  - Different predators
  - Different climate
- Eventually, they become **so genetically different** they can no longer interbreed → **new species formed**.

📌 **Classic Example — Darwin's Finches (Galápagos Islands):**

- Finches from South America migrated to various islands.
- Each island provided different types of food sources.
- Over time, beak shapes evolved into distinct forms such as seed-crackers, insect-pickers, and cactus-feeders.
- Currently, approximately 15 different finch species exist, all originating from a single common ancestor.

📌 **Indian Example — Squirrels of Western Ghats:**

- Different squirrel populations were separated by valleys.
- These populations evolved distinct color patterns.
- The squirrels also evolved different sizes.

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2. 🔬 Genetic Variation (Mutation & Natural Selection)

- **Random mutations** occur in DNA during reproduction.
- Some mutations are **advantageous** in a given environment → those individuals **survive and reproduce more** (Natural Selection).
- Over many generations, accumulated genetic changes make the population **genetically incompatible** with the ancestor.

📌 **Example — Antibiotic-Resistant Bacteria:**

- A random mutation occurs in bacteria, allowing some individuals to survive antibiotic exposure.
- These resistant bacteria then reproduce.
- Over time, the resulting population becomes sufficiently distinct to be classified as a new strain or variant.

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3. 🚫 Reproductive Isolation (Sympatric Speciation)

- Two populations **living in the same area** but becoming reproductively isolated due to:
  - **Different mating seasons** (temporal isolation)
  - **Different mating behaviors** (behavioral isolation)
  - **Genetic incompatibility** (gametic isolation)
- No physical barrier needed — isolation is **biological**.

📌 **Example — Apple Maggot Fly (*Rhagoletis pomonella*):**

- The species originally lived on hawthorn berries.
- With the introduction of apple trees to North America, a segment of the fly population switched to apples.
- Apple-preferring flies now exclusively mate on apple trees, while hawthorn flies mate on hawthorn trees.



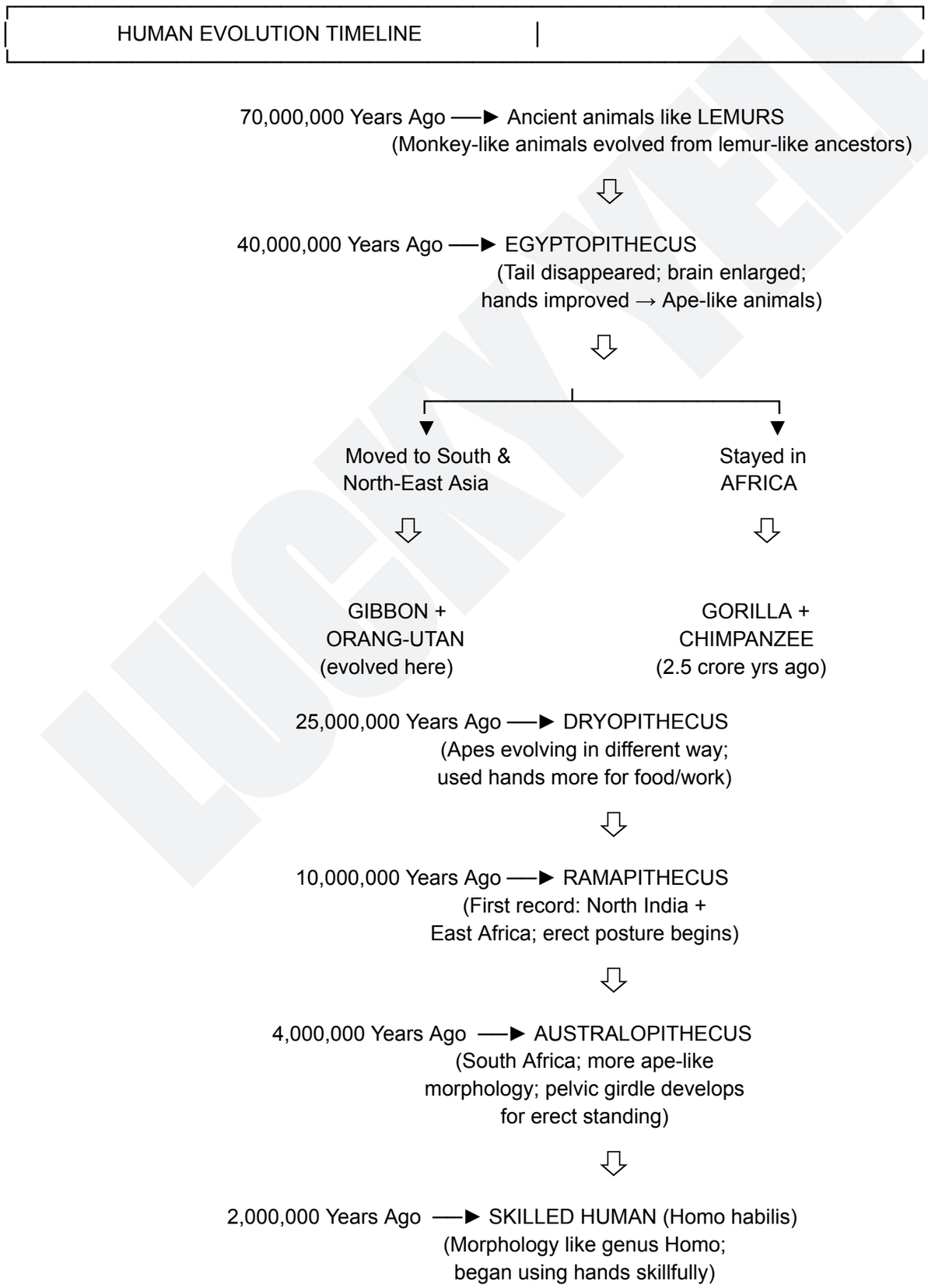
- This behavioral change has led to the formation of two distinct populations within the same geographical region.

📌 **Example — Flowering Time in Plants:**

- Two populations of the same wildflower species exhibit different blooming periods, with one blooming in April and the other in June.
- The difference in timing prevents the exchange of pollen between the two groups.
- This isolation leads to gradual genetic divergence over time.
- The eventual result of this process is the formation of a new species.

🦴 9. Human Evolution

🕰️ Evolutionary Timeline of Humans



⇩

1,500,000 Years Ago ⟶ MAN WITH ERECT POSTURE
(Homo erectus; existed in
China and Indonesia)

⇩

Brain developed over ~1 lakh years
Fire discovered

⇩

150,000 Years Ago ⟶ NEANDERTHAL MAN
(First example of Homo sapiens
"wise man")

⇩

50,000 Years Ago ⟶ CRO-MAGNON MAN
(Homo sapiens sapiens;
evolution accelerated)

⇩

10,000 Years Ago ⟶ Agriculture practiced;
Cattle rearing; Cities established

⇩

5,000 Years Ago ⟶ Art of writing invented
→ History begins

⇩

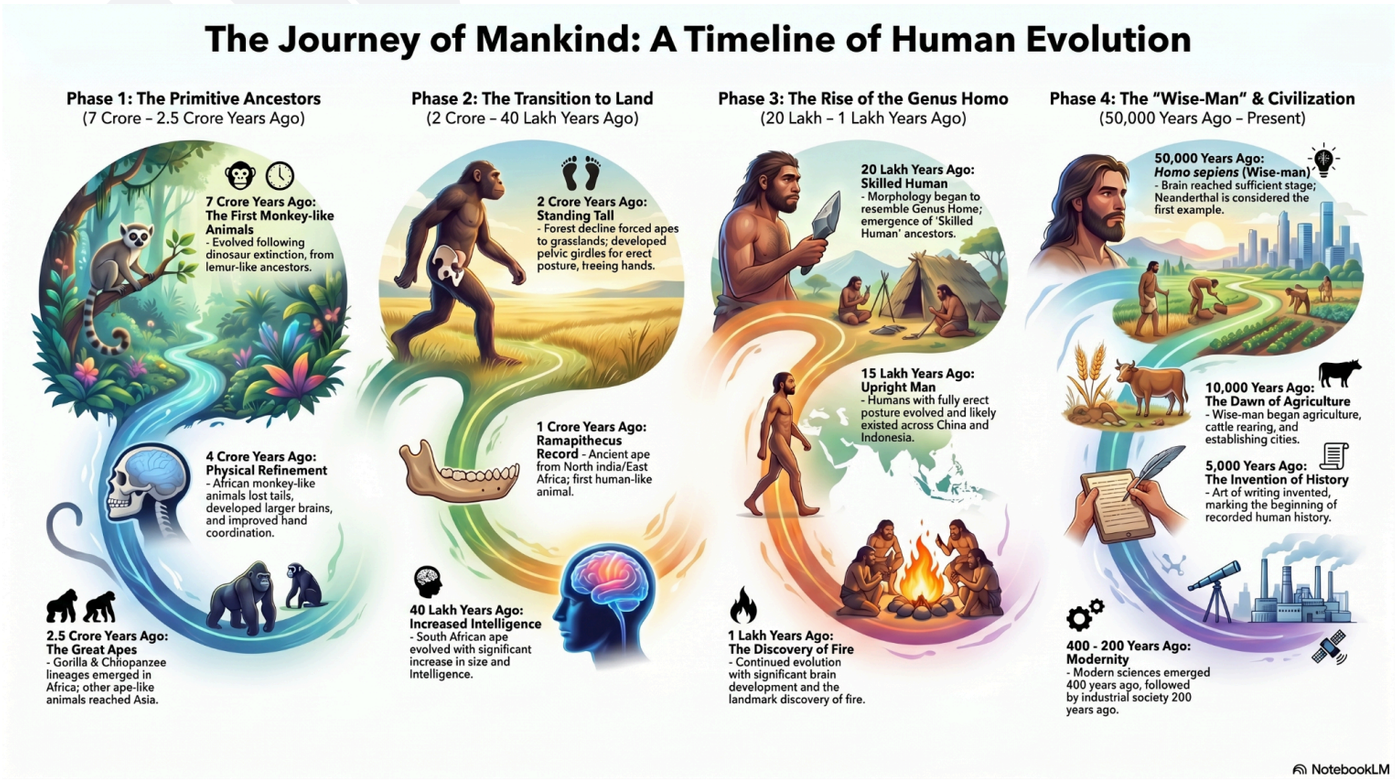
400 Years Ago ⟶ Modern sciences emerged

⇩

200 Years Ago ⟶ Industrial society established

⇩

MODERN HUMAN 🧑



📊 Key Milestones in Human Evolution Table

Ancestor	Years Ago	Key Development
Lemurs	7,00,00,000	Ancient primate ancestors
Egyptopithecus	4,00,00,000	Tail lost; brain enlarged
Dryopithecus	2,50,00,000	Hands used more for food
Ramapithecus	1,00,00,000	First human-like, North India + East Africa
Australopithecus	40,00,000	Erect posture in grasslands
Skilled Human	20,00,000	Genus Homo morphology
Erect Man	15,00,000	Asia (China, Indonesia)
Neanderthal	1,50,000	First Homo sapiens; fire discovered
Cro-Magnon	50,000	Rapid evolution; fully modern brain
Agriculture begins	10,000	Cities; cattle rearing
Writing invented	5,000	History begins

🎯 Key Takeaways

Sr. No.	Concept	Core Idea
1	**Heredity**	Transfer of biological characters via genes from one generation to the next
2	**Central Dogma**	DNA → mRNA → Protein (Transcription followed by Translation)
3	**Transcription**	mRNA synthesized from one strand of DNA; uracil replaces thymine
4	**Translation**	mRNA codons (triplets) decoded by tRNA anticodons into an amino acid chain
5	**Translocation**	Ribosome moves along mRNA one triplet codon at a time
6	**Mutation**	Sudden change in position or sequence of a nucleotide in a gene
7	**Evolution**	Gradual change in organisms over long periods leading to new species
8	**Morphological Evidence**	Similar external features indicating common ancestry
9	**Anatomical Evidence**	Similar internal bone structures (homologous organs) across different animals
10	**Vestigial Organs**	Degenerated useless organs (e.g., appendix) as proof of evolution
11	**Carbon Dating**	Ratio of C-14 to C-12 determines the age of fossils
12	**Connecting Links**	Organisms with characters of two different groups (e.g., Duck-billed Platypus)

Sr. No.	Concept	Core Idea
13	**Embryological Evidence**	Extreme similarities in early vertebrate embryos indicate common origin
14	**Darwinism**	Theory of Natural Selection and Survival of the Fittest
15	**Lamarckism**	Theory of inheritance of acquired characters through use or disuse of organs
16	**Speciation**	Formation of new species via genetic variation and isolation
17	**Human Evolution**	7-crore-year journey from lemur-like ancestors to modern Homo sapiens

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💡 **Exam Tip:** For 5-mark questions on **evidences of evolution**, use the mnemonic **"MAVPCE"** — **M**orphological, **A**natomical, **V**estigial, **P**alaeontological, **C**onnecting links, **E**mbryological.

